Jiaqi Wu

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Summary

UC San Diego dual-degree candidate in Mathematics–Computer Science and Cognitive Science–Machine Learning and Neural Computation, graduating in December 2026 with plans to begin full-time M.S. study in Fall 2027. Research experience spans large-scale distributed machine learning, biomedical foundation models, autoregressive image generation, and multi-agent LLM systems.

Education

University of California San Diego — La Jolla, CA
B.S. in Mathematics–Computer Science — Dec 2026
B.S. in Cognitive Science–Machine Learning and Neural Computation
GPA: 3.96/4.00.
Relevant Coursework: Deep Machine Learning, Reinforcement Learning, Optimization, Numerical Analysis, Recommender Systems, Data Mining

Experience

Research Assistant Intern — Apr 2026 – Present
mlPC Lab, UCSD — *La Jolla, CA*
- Lead post-training of Qwen-Image-Edit for controllable image generation and editing using supervised fine-tuning and reinforcement learning, targeting instruction adherence and reliable edits.
- Build distributed PyTorch training and evaluation pipelines with Hugging Face Accelerate and FSDP; analyze model behavior through controlled ablations and quantitative evaluation of edit fidelity, content preservation, and controllability.

Research Assistant Intern — Sep 2025 – July 2026
ACM Lab, UNC Chapel Hill — *Remote*
- Architected and pretrained a Transformer foundation model for biomedical time-series at population scale (100K+ resting-state fMRI scans) via a distributed PyTorch training pipeline.
- Evaluated transferability via linear probing and fine-tuning on downstream clinical prediction tasks, outperforming task-specific baselines.

MATLAB Tutor — Mar 2025 – Dec 2025
Department of Mathematics, UCSD — *La Jolla, CA*
- Tutored 50+ undergraduate students in MATLAB, linear algebra, and differential equations through one-on-one debugging, conceptual review, and programming support.

Full-Stack Developer — Jul 2024 – Oct 2024
UCSD CSSA — *La Jolla, CA*
- Prototyped a mobile community-engagement app and implemented backend data models, authentication, and client-server communication using Node.js and Tencent Cloud.

Projects

LLM Course Planning System — Jan 2025 – Feb 2026
Swift, AWS, LLM Agents
- Built a multi-agent GraphRAG pipeline that transforms university catalogs into course knowledge graphs, enabling prerequisite-aware retrieval and personalized academic planning.
- Developed a native SwiftUI iOS app and AWS serverless backend with Lambda, API Gateway, DynamoDB, and S3 for interactive degree roadmaps and multi-term scheduling.

Awards & Honors

- SanD Hackathon – Multi-Agent System Challenge Winner, San Diego
- Valedictorian
- AMC 12 Certificate of Distinction
- MATHCON Finals, Honorable Mention, Chicago

Skills

Languages: Python, C/C++, JavaScript, Swift, HTML/CSS
ML: PyTorch, NumPy, pandas, scikit-learn, Transformers, representation learning
Tools/Cloud: Git, Linux, Claude Code, AWS Lambda, S3, DynamoDB, Xcode, MATLAB, LaTeX